

NOLINSK, Russian Federation

WMO: 273930

Lat: **57.550N**Lon: **49.950E**Elev: **134**StdP: **99.72**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.5	-26.1	-32.7	0.2	-29.4	-29.0	0.3	-25.9	9.4	-5.6	8.6	-5.6	1.8	0	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	29.4	20.3	27.5	19.2	25.8	18.3	21.2	27.4	20.3	26.1	19.3	24.4	2.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.1	24.0	18.2	13.3	22.9	17.2	12.5	22.2	61.9	27.6	58.9	26.6	55.4	24.4	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.5	6.4	DB	-32.7	32.0	3.9	2.0	-35.4	33.4	-37.7	34.5	-39.8	35.7	-42.6	37.1	(4)
(5)			WB	-32.8	22.8	3.8	1.3	-35.5	23.7	-37.8	24.4	-39.9	25.1	-42.7	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-12.1	-11.4	-4.7	4.0	11.8	16.4	19.0	16.4	10.6	3.8	-4.1	-9.9
	DBStd	12.31	7.44	7.26	4.69	5.01	5.19	4.43	3.78	4.26	4.26	4.29	6.04	7.48
	HDD10.0	3261	684	601	456	191	43	5	0	2	43	198	424	616
	HDD18.3	5581	943	834	714	431	211	88	38	90	233	452	674	874
	CDD10.0	849	0	0	0	11	98	196	278	201	62	5	0	0
	CDD18.3	127	0	0	0	0	7	29	58	30	2	0	0	0
	CDH23.3	1052	0	0	0	1	77	241	477	244	12	0	0	0
	CDH26.7	262	0	0	0	0	12	47	126	77	1	0	0	0
Wind	WSAvg	2.9	3.1	3.1	3.3	2.9	2.9	2.6	2.2	2.4	2.6	3.1	3.3	3.3
Precipitation	PrecAvg	632	47	33	39	34	47	65	75	64	55	70	55	48
	PrecMax	768	85	78	80	74	101	147	139	169	127	196	114	77
	PrecMin	484	15	6	11	8	11	19	11	12	10	18	12	16
	PrecStd	75	18	17	16	19	24	28	37	36	27	34	24	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	1.7	7.1	22.0	28.5	30.4	32.4	32.3	25.0	17.7	8.6	3.9
		MCWB	0.4	0.6	3.7	12.8	18.0	20.5	20.5	20.5	16.6	11.5	7.7	3.6
	2%	DB	0.4	0.7	4.4	18.1	25.5	27.8	29.6	28.7	22.1	13.8	5.8	1.7
		MCWB	-0.2	0.1	1.5	10.5	15.8	19.2	20.6	20.3	15.4	10.0	5.0	1.1
	5%	DB	-0.6	0.1	2.4	14.7	23.1	26.1	27.9	25.8	19.8	11.5	3.8	0.6
		MCWB	-1.1	-0.3	0.5	8.9	14.5	18.4	19.7	18.7	14.4	9.0	2.9	0.1
	10%	DB	-2.3	-1.4	1.3	11.8	20.6	24.2	26.1	23.5	17.5	9.9	2.2	-0.4
		MCWB	-2.8	-2.1	0.0	7.0	13.1	17.5	19.0	17.6	13.4	8.3	1.5	-0.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	1.0	3.9	13.4	19.1	22.1	22.6	21.7	18.1	12.7	7.8	3.6
		MCDB	1.3	1.4	6.9	20.1	26.8	28.1	28.2	28.7	22.7	15.5	8.4	3.8
	2%	WB	0.0	0.3	2.0	11.3	16.9	20.5	21.4	20.6	16.3	10.8	5.1	1.2
		MCDB	0.4	0.7	3.6	16.6	23.3	26.2	27.4	27.3	20.5	13.0	5.6	1.6
	5%	WB	-1.1	-0.3	1.0	9.3	15.4	19.3	20.6	19.3	14.9	9.4	3.2	0.2
		MCDB	-0.5	0.1	2.0	14.3	21.2	24.8	26.5	24.7	18.7	11.1	3.7	0.6
	10%	WB	-2.8	-2.1	0.3	7.4	13.9	18.1	19.7	18.1	13.7	8.3	1.6	-0.9
		MCDB	-2.3	-1.5	1.3	11.4	19.3	23.0	24.9	22.6	17.3	9.8	2.0	-0.4
Mean Daily Temperature Range	5% DB	MDBR	5.7	6.6	6.8	8.3	10.0	9.3	9.2	8.6	7.7	5.2	4.1	5.0
		MCDBR	4.7	5.1	7.0	12.8	13.3	11.4	12.0	11.9	11.5	8.4	4.5	3.5
		MCWBR	4.7	4.7	5.2	6.8	6.1	5.3	5.1	4.8	6.0	5.5	4.4	3.7
	5% WB	MCDBR	4.5	4.9	5.5	12.1	11.9	10.4	10.7	10.7	10.4	7.5	4.4	3.5
		MCWBR	4.5	4.8	4.5	6.8	6.2	5.4	5.2	4.9	6.0	5.5	4.4	3.7
Clear-Sky Solar Irradiance	taub	0.284	0.312	0.360	0.395	0.379	0.374	0.399	0.411	0.365	0.344	0.303	0.275	
	taud	2.037	2.048	1.984	2.099	2.344	2.397	2.337	2.283	2.387	2.316	2.155	2.010	
	Ebn at Noon	535	698	762	807	856	865	830	788	775	674	518	421	
	Edh at Noon	59	91	128	135	113	109	113	112	87	71	52	45	
All-Sky Solar Radiation	RadAvg	0.54	1.42	2.91	4.22	5.34	5.67	5.61	4.26	2.56	1.11	0.45	0.30	
	RadStd	0.07	0.24	0.30	0.43	0.48	0.42	0.53	0.42	0.30	0.18	0.07	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page